Title: Automated Detection of Potato Leaf Diseases Using Machine Learning-Based Image Analysis

Presented by

Mannan Hossain
221-15-5176
Dhrubo Chandra Mondal
221-15-5329
Department of CSE
Daffodil International University

Supervised by

Rahmatul Kabir Rasel Sarker
Lecturer
Department of CSE
Daffodil International University

Objective

Our main goal is to develop a machine learning-based tool to detect and classify potato leaf diseases accurately using image analysis

Automate disease detection from leaf images

Improve early diagnosis for better crop management

Enhance agricultural productivity using AI

cont..

Motivation

Potato leaf diseases pose a significant threat to crop yield and quality.

Key Issues:

- Early detection is challenging for farmers
- Limited automated solutions available
- Traditional methods are time-consuming and less accurate
- Why It Matters?

- Disease outbreaks can lead to significant crop losses
- Lack of efficient detection methods affects agricultural productivity
- AI-powered detection can revolutionize plant disease management

cont..

Background Study

1. Potato Leaf Diseases & Impact – Potato crops suffer from various diseases that affect yield and quality. This study focuses on four major diseases: Early Blight, Late Blight, Leaf Roll Virus, and Mosaic Virus, along with healthy leaves for comparison.

2. Symptoms & Causes – Each disease has distinct symptoms: fungal infections (Early & Late Blight) cause dark lesions, while viral infections (Leaf Roll & Mosaic Virus) lead to leaf deformities and discoloration.

3. Detection & Classification – Traditional methods rely on visual inspection, but modern techniques like image processing and machine learning provide faster and more accurate disease identification.

cont..

Gap Analysis

Current Challenges in Potato Leaf Disease Detection:

- Traditional methods rely on manual inspection, which is time-consuming and error-prone
- Limited research on AI-based automated detection systems
- Existing models lack high accuracy in real-world agricultural conditions
- Research Gap
- A need for a robust, AI-driven system for accurate and early disease detection

cont..

Methodology

01 Dataset Collection

02 Image Preprocessing

03 Model Selection & Training

04 Validation & Testing

05 Deployment

Expected Outcomes

- A high-accuracy machine learning model for potato leaf disease detection
- Faster and more reliable identification compared to manual methods
- A user-friendly tool that can assist farmers in early disease detection
- Contribution to AI-driven solutions in agriculture

cont..

Conclusion

Potato leaf diseases pose a significant threat to crop production, making early detection essential for effective disease management. This study focuses on distinguishing between healthy and diseased leaves using advanced image processing techniques. By leveraging modern technology, we can develop a more efficient and automated approach to disease identification, ultimately contributing to higher yields and sustainable potato farming.

cont..

THANK YOU